Samiur R. Mir

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Education

Oklahoma State University — Oklahoma, United States
Ph.D. in Physics — *Aug. 2021 – Present*
Advisor: Prof. Vedran Brdar

University of Alberta — Alberta, Canada
M.Sc. in Physics — *Sept. 2019 – Apr. 2021*

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Physics — *Jan. 2015 – Aug. 2019*

Publications & Preprints

6. Vedran Brdar, **Samiur R. Mir**, Xun-Jie Xu, *"Solar Constraints on Heavy Neutral Leptons with ν_τ Mixing"*, arXiv:2608.19319 [hep-ph] (under PRL review)

5. Vedran Brdar, **Samiur R. Mir**, *"Ultra-High-Energy Tau Neutrinos as Probes of Lorentz Invariance"*, JCAP 11 (2026) 008, arXiv:2604.19880 [hep-ph]

4. Vedran Brdar, Dibya S. Chattopadhyay, **Samiur R. Mir**, Tousif Raza, and Marc S. Romanowski, *"Prospects for Neutrino Observation and Mass Measurement from Binary Neutron Star Mergers"*, Phys. Rev. D 113, 123017 (2026), arXiv:2511.16658 [hep-ph]

3. **Samiur R. Mir**, Carlos A. Argüelles, K.S. Babu, Vedran Brdar, *"Double Bangs in IceCube as a Window to the Neutrino Mass Origin"*, Phys. Lett. B 875 (2026) 140322, arXiv:2511.07541 [hep-ph]

2. Vedran Brdar, Ying-Ying Li, **Samiur R. Mir**, Yi-Lin Wang, *"Collider Prospects for the Neutrino Magnetic Moment Portal"*, JHEP 10 (2025) 230, arXiv:2502.07024 [hep-ph]

1. M. Jamil Aslam, Wen Chen, Andrzej Czarnecki, **Samiur R. Mir**, Muhammad Mubasher, *"Rare Decays of the Positronium Ion and Molecule,* $\mathrm{Ps}^- \to e^-\gamma$ *and* $\mathrm{Ps}_2 \to e^+e^-\gamma$, $\gamma\gamma$, e^+e^-*"*, Phys. Rev. A 104, 052803 (2021), arXiv:2108.06785 [hep-ph]

Academic Honors and Awards

Distinguished Graduate Fellowship | *OSU Foundation and the Graduate College* 2026
Student Travel Awards | *granted by APS Fall Meeting* 2025
Student Travel Awards | *granted by APS and Pheno organizers* 2024
Recipient of Honours Scholarship (top 5% CGPA in Bachelors) | *University of Dhaka* 2019

Seminar Talks

Heavy Neutral Leptons from the Sun: New Limits on Mixing with Tau Neutrinos 2026
Neutrino Theory Community seminar, US (online)

Heavy Neutral Leptons from the Sun: New Limits on Mixing with Tau Neutrinos 2026
CNP seminar, Center for Neutrino Physics, Virginia Tech

RG-Running Neutrino Mixing at IceCube 2026
Latin American Webinars in Physics (Webinar 195, available on YouTube)

Probing Neutrinophilic BSM Scenarios with High-Energy Neutrinos 2026
HEP seminar, PITT PACC, University of Pittsburgh

Unveiling the Neutrino Magnetic Moment: Constraints from Colliders 2025
LPPC seminar, Harvard University

Conference Presentations

Testing Lorentz Invariance with Ultra-High-Energy Tau Neutrinos 2026
SUSY Conference, University of Oklahoma

Testing RG Evolution of Neutrino Mixing with IceCube Double Bang Events 2026
Pheno Symposium, University of Pittsburgh

Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube 2025
Particle Physics on the Plains, University of Kansas

Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube 2025
APS Meeting Texas, Arkansas, Oklahoma Section, Texas Christian University

Unveiling the Neutrino Magnetic Moment: Constraints from Colliders 2025
Pheno Symposium, University of Pittsburgh

Unveiling the Neutrino Magnetic Moment: Constraints from Colliders 2025
LPPC seminar at Harvard University

Neutrino Magnetic Moment at Colliders 2024
Particle Physics on the Plains, University of Kansas

Collider Probes of Neutrino Magnetic Moment 2024
DPF - Pheno Symposium, University of Pittsburgh

Single Photon Decay of the Positronium Ion 2020
57th Winter Nuclear and Particle Physics Conference, Banff, Alberta

How to Find Decay Distribution from Symmetry 2019
10th Annual Symposium for Graduate Physics Research, University of Alberta

Mentoring Experience

Safius Sakib Shuddho, University of Dhaka Aug. 2026 – Present
ICTP PWF Physics For Bangladesh Internship

Marc Romanowski, The College of New Jersey June 2025–Aug. 2025
NSF REU Visiting Undergraduate Researcher

Summer Schools

Cosmic Neutrino Practicum | *Harvard University* June 2026
International Neutrino Summer School | *Fermilab* Aug. 2025

Leadership Activities

Coordinator and Contributor, HEP-pheno Student Seminar 2023 – 2025
Dept. of Physics, Oklahoma State University
Vice-President, Physics Graduate Student Association 2023 – 2025
Dept. of Physics, Oklahoma State University
Outreach Volunteer, Physics Open House 2023 – 2025
Dept. of Physics, Oklahoma State University
Academic Team Member | *Bangladesh Mathematical Olympiad* 2015 – 2019
Academic Instructor | *Community of Physics, Bangladesh* 2015 – 2019

Teaching Experience

Oklahoma State University
Teaching Assistant: Electricity & Magnetism (Labs & Discussions) Aug. 2021 – Present
University of Alberta
Teaching Assistant: Particles and Waves, Electricity and Magnetism (Labs) Aug. 2019–April 2021

Professional References

- **Prof. Vedran Brdar** (Ph.D. Supervisor)
  Assistant Professor
  Department of Physics
  Oklahoma State University
  Email: vedran.brdar@okstate.edu
- **Prof. Carlos Argüelles-Delgado**
  Assistant Professor
  Department of Physics
  Harvard University
  Email: carguelles@g.harvard.edu
- **Prof. Kaladi S. Babu**
  Regents Professor of Physics
  Department of Physics
  Oklahoma State University
  Email: kaladi.babu@okstate.edu